

THE ZODIAC

ALTHOUGH IT IS NOT OBVIOUS BECAUSE OF THE SUN'S GLARE, AS EARTH ORBITS THE SUN, THE SUN SEEMS TO MOVE AGAINST THE BACKDROP OF STARS, ALWAYS STAYING WITHIN A BAND OF THE CELESTIAL SPHERE CALLED THE ZODIAC.

During the course of this annual journey around the celestial sphere, the Sun moves along a circle called the ecliptic (see p.90). An imaginary band around the celestial sphere that extends for about 8–9° on either side of the ecliptic is called the zodiac. The ecliptic passes through 13 constellations that lie, at least in part, in the zodiac and these are known as zodiacal constellations. The astrological zodiac is divided into 12 equal segments, called “signs,” and excludes the constellation Ophiuchus.

The Sun spends a period of time in each zodiacal constellation but the dates it does so do not correspond with those ascribed

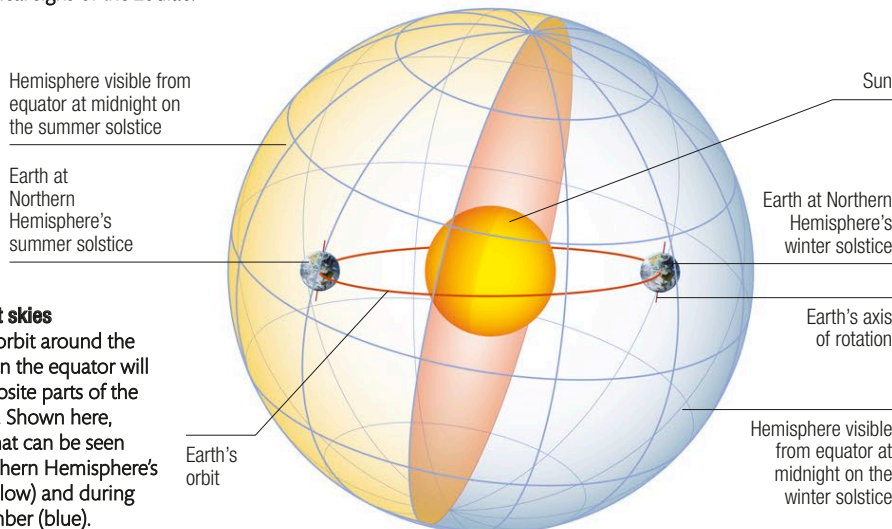
to the astrological signs. This is due to the effects of precession and because the constellations are not all the same size.

Whenever the Sun is moving through a particular area of the zodiac, the stars in that part of the celestial sphere cannot be seen because of the glare. Rather, the most easily observed parts of the celestial sphere are always those on the opposite side to Earth from the Sun. These are the parts visible in the middle of the night. Over the course of a year, as Earth orbits the Sun, the portions of the celestial sphere—including the different parts of the zodiac—that can be viewed from Earth at night quite dramatically alter.

SUN'S PROGRESS			
Constellation	Dates in each constellation	Constellation	Dates in each constellation
Aries	April 19 – May 13	Scorpio	November 23 – 29
Taurus	May 14 – June 19	Ophiuchus	November 30 – December 17
Gemini	June 20 – July 20	Sagittarius	December 18 – January 18
Cancer	July 21 – August 9	Capricorn	January 19 – February 15
Leo	August 10 – September 15	Aquarius	February 16 – March 11
Virgo	September 16 – October 30	Pisces	March 12 – April 18
Libra	October 31 – November 22		

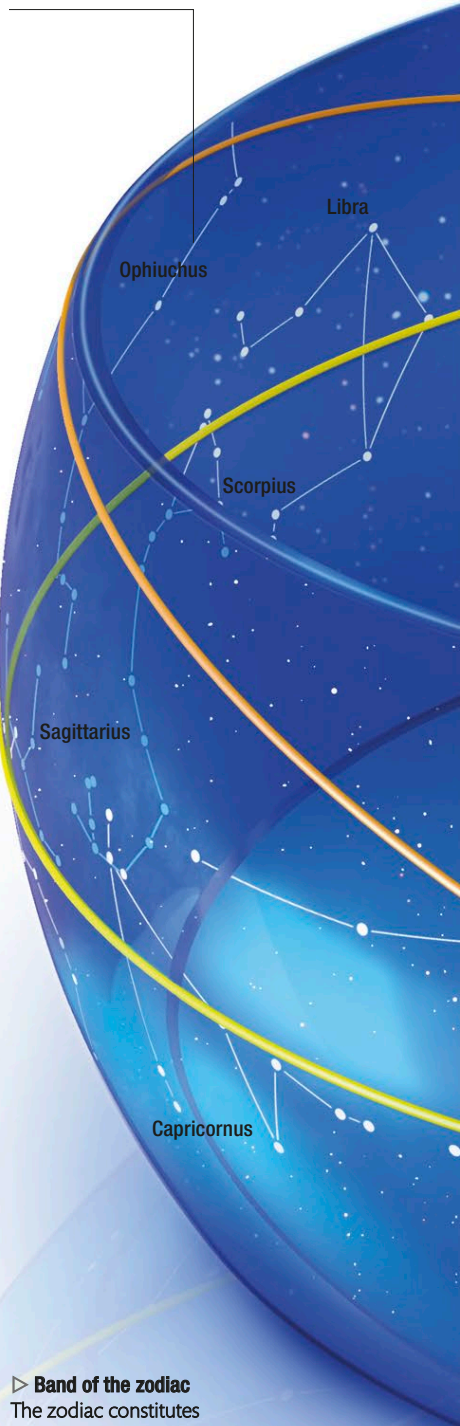
△ **Days of the zodiac**
The dates the Sun passes through the 13 zodiacal constellations are completely different from the dates associated with the astrological signs of the zodiac.

▷ **June and December night skies**
At opposite sides of Earth's orbit around the Sun, an observer standing on the equator will be able to view exactly opposite parts of the celestial sphere at midnight. Shown here, for example, are the parts that can be seen at midnight during the Northern Hemisphere's summer solstice in June (yellow) and during the winter solstice in December (blue).



Ophiuchus, the 13th constellation of the zodiac

Winter solstice in Northern Hemisphere, the point where the Sun is farthest below the celestial equator



▷ **Band of the zodiac**
The zodiac constitutes about one-sixth of the surface area of the celestial sphere (its depth is exaggerated here). The ecliptic runs through its center. As well as the Sun, the celestial paths of the Moon and the planets of the Solar System are also restricted to the zodiac.